# User needs analysis to design a 3D multimodal protein-docking interface

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**ABSTRACT**

Protein-Protein docking is a recent practice in biological research which involves using 3D models of proteins to predict the structure of complexes formed by these proteins. Studying protein-protein interactions and how proteins form molecular complexes allows researchers to better understand their function in the cell. Currently, the most common methods used for docking are fully computational approaches, followed by the use of molecular visualization tools to evaluate results. However, these approaches are time consuming and provide a large number of potential solutions. Our basic hypothesis is that a virtual reality (VR) framework for molecular docking can combine the benefits of multimodal rendering, of the biologist's expertise in the field of docking, and of automated docking algorithms. We think this approach will increase efficiency in reaching the solution of a docking problem. However designing immersive and multimodal virtual environments (VE) based on VR technology calls for clear and early identification of user needs. To this end, we have analyzed the task of protein-protein docking as it is carried out today, in order to identify benefits and shortcomings of existing tools, and support the design of new interactive paradigms. Using these results, we have defined a new approach and designed a multimodal application for molecular docking in a virtual reality context.

**KEYWORDS:** Protein-protein docking, User centered design, Virtual Reality, Multimodal rendering.

**INDEX TERMS:** H.5.2 [User Interfaces]: Ergonomics, User-centered design, Graphical user interfaces (GUI), Auditory feedback, Voice I/O, Auditory (non-speech) feedback, Haptic I/O. H.5.1 [Multimedia Information Systems]: Artificial, augmented, and virtual realities, Audio input/output, Evaluation/methodology. J.3 [Life and Medical Sciences]: Biology and genetics

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## 1 INTRODUCTION

Proteins are large organic compounds made of amino acids arranged in a linear chain and joined together by peptide bonds between the carboxyl and amino groups of adjacent amino acid residues. The sequence of amino acids in a protein is defined by a gene and encoded in the genetic code. Although this genetic code specifies 20 standard amino-acids, residues in a protein are often chemically altered before the protein can acquire a 3D structure and a function in the cell. Like other biological macromolecules, proteins are essential parts of living organisms and participate in every process within cells. Proteins also have structural or mechanical functions, work together to achieve specific functions and often associate to form stable complexes (see Figure 1).

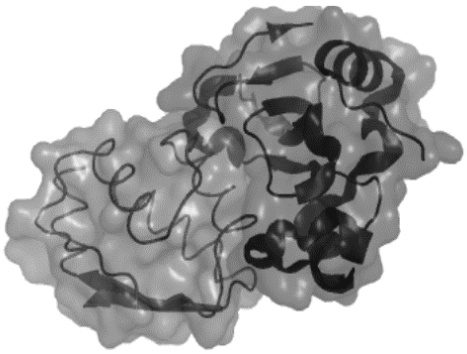


Figure 1. The Barnase (left) /Barstar (right) complex: surfaces (transparency) and ribbons representations

Protein-protein docking involves using 3D protein models obtained by crystallography or Nuclear Magnetic Resonance. The goal is to understand which proteins can form stable complexes, how these complexes are formed, and what their role in the cell is.

During complex formation, biologists have to take into account physicochemical interactions. The most significant are hydrophobic, electrostatic and Van der Waals forces. Moreover, they have to consider bonds between hydrogen or sulphur atoms on the interface of the two proteins. Automatic tools have been developed to partially solve the docking problem and help the user in this complex task. However, as shown in the first part of this paper, fully computational approaches are not yet efficient enough to provide accurate and definite results.

For this reason, it is important to investigate another approach to docking. We think this approach will increase efficiency in reaching the solution of a docking problem.

However, designing immersive and multimodal virtual environments (VEs) calls for a clear and early identification of user needs. To this end, we conducted an ergonomic analysis of the task of protein-protein docking as it is carried out today, in order to identify benefits and shortcomings of existing tools, as well as to support the design of new interactive paradigms. Section 3 presents the methodology we used, the user needs we elicited, and ergonomic recommendations for designing a

functional prototype of an immersive and multimodal virtual environment for molecular docking.

Finally, section 4 details our VR framework, integrated into the docking practices of biologists. Technical specifications and development choices for designing the prototype, to respect existing constraints and users requirements, are described.

## 2 Protein-Protein Docking in Silico: A short State of the Art

### 2.1 Automated Protein-Protein Docking

Most computational approaches of docking can be divided into three successive stages: representation, search and evaluation.

Many algorithms dealing with the search for protein binding sites are based on an exhaustive approach. They explore the surface of each molecule in order to scan all potential solutions [1]. Indeed, docking is based, first of all, on the complementarity of protein surfaces. The main problem with these approaches is combinatorial explosion. Some sampling procedures, such as genetic algorithms [21], or Monte Carlo simulations [16] have also been tested for docking problems. Yet they do not necessarily produce correct solutions, and computing time remains long (from a few minutes to several hours).

The protein configurations provided by the previous step, must then be evaluated by a scoring function. This function evaluates interaction between proteins. Therefore, the accuracy of the scoring function also has great impact on the quality of molecular docking results. Existing scoring functions can be classified in two main categories [29]: force-field based scoring functions and empirical scoring functions. Force field based functions only consider a few interaction terms, related to electrostatic properties. Empirical functions aim to reproduce experimental values of binding energies and reach the global minimum for experimentally examined complex configurations.

Therefore, automatic processes must overcome three difficulties. The first is to deal with a very large search space, because of the large number of results and potential binding sites for any given pair of molecules. As a consequence, it could not be covered completely in a suitable time. The second problem is due to the search algorithms. These are based on a minimum search scheme and often produce a local minimum. Finally, the last drawback is accuracy. While docking yields conclusive results in many cases, the risk for selecting false positives is high. The result of the process provides a large set of possible solutions, many of which are not biologically valid [29] and cannot all be experimentally tested.

In order to address all these problems associated to the fully automatic approach, researchers use molecular visualization tools to help the docking task. However, it is quite difficult to simultaneously manipulate two 3D structures in a classical desktop interaction paradigm. VR seems, therefore, to be more adapted to the docking task, because this technology can easily allow direct interaction, and adaptative stereovision may be more suitable to visualize and understand 3D structures such as proteins. The following part presents a short state of the art of docking applications in VEs.

### 2.2 VR and multimodal interaction for Protein-Protein Docking

Taking into account the limits of the traditional proteins docking algorithms (see. section 2.1) and new possibilities offered by virtual reality, the use of VR to support molecular docking has gathered some interest in past years. Early work focused mainly on technical needs and limitations. For example, STALK [18] used parallel distributed processing for graphical rendering of 3D molecular models and algorithmic generation of possible solutions. With STALK, users could visualize two proteins, attend the progress of the algorithm, suspend it in order to move one of the molecules using a 3D mouse, and resume its execution with the new position. Evaluations were performed by comparing binding energy of the STALK system vs. automatically generated complexes and showed that configurations docked manually using STALK were inferior.

Later efforts such as VRDD [2] implemented a Monte Carlo local search algorithm allowing refinement through exploration of similar solutions. Evaluations were carried out on three test cases, by comparing RMSD (Root Mean Square Deviation) between a docking solution and the crystallographic structure of the same complex. Results were deemed conclusive for two cases, but not for the third case, which involved (1) large proteins, i.e. a large number of possible solutions, (2) flexible proteins which undergo conformational changes upon binding and (3) a "counterintuitive" binding mechanism. If multimodal VEs are to be used as an alternative to existing graphics-based tools, they need to yield similar or superior results even for such difficult problems. Therefore, such work allowed identification of technical issues, e.g. real time computation of binding energy, constructing flexible molecular models, etc. Recent work, e.g. [24], has focused on solving these issues at the expense of "natural interaction", by producing "enhanced visualization" plug-ins compatible with mainstream molecular visualization software such as VMD [29].

Many projects have also explored the potential of multimodal rendering to support molecular docking. A first point is that such projects often use haptic or auditory feedback, rarely both.

Haptic-oriented projects are mainly geared toward providing force feedback constraints to guide the user toward the best solution. The use of other types of haptic feedback (e.g. vibrotactile) is almost nonexistent. These projects include GROPE [8], IVPS [19], and SenSitus, a haptic plug-in for VMD [29] and involve user exploration of molecular surfaces with a PHANToM™ device. These projects aim to provide means for simultaneous rendering of multiple volumetric datasets and to test various interactive paradigms. The software allowed moving molecules in relation to one another as well as haptic feedback on collision detection. Evaluation was based on measuring docking errors and completion time for various interactive paradigms (e.g. "hard" vs. "soft" docking, monoscopic vs. stereoscopic visualization). Visual stereoscopy was found to reduce both error rate and completion time, but haptic feedback increased completion time due to the necessity for local, sequential exploration.

Audio-oriented projects aim to provide the user with clues about molecular properties, in particular binding properties, using earcons (abstract sounds, e.g. a pure tone), auditory icons (everyday sounds) or sonification (mapping of sound parameters in relation to scientific data) [14]. Early work involved sonification of sequential data, e.g. DNA or amino acids sequences. In contrast, docking involves processing data based on interaction between several hundreds of atoms, and applications which use sonification to study biomolecules are rare.

In sum, beyond purely technical limitations, lack of knowledge regarding user requirements produces a gap between outcomes envisioned for such projects (i.e. natural interaction with CG proteins allowing quick and easy construction of molecular complexes). Such projects usually yielded technically impressive systems, that are however unusable as work tools. Besides its technological innovations, our project's novelty rests in an early involvement of users in the design process which should help us to avoid such pitfalls. This approach is expected, (1) to accelerate and facilitate the development of functional prototypes through relevant choices in design, (2) to improve user acceptance of these

novel interfaces, thereby encouraging further development and spreading of our applications, and (3) to suggest unexpected new directions for research and innovation [1].

In conclusion, various works have been carried out regarding immersive docking, but the results obtained are still too specific or not convincing enough. Besides, they place the user more in the role of an observer or controller of an automatic process than in the role of an actor. Our approach is more human-centred and addresses the design of a system for immersive interaction to carry out protein-protein docking. Our main goal was to determine design principles in order to integrate VE interactions in everyday practice of docking tasks as performed by researchers in biochemistry. To this end, we present in the next section the analysis of the task of protein-protein docking as carried out today, in order to identify benefits and shortcomings of existing tools, as well as to support the design of new interactive paradigms.

## 3 User Need Analysis To Support Design Of Multimodal Docking VE

### 3.1 Data Collection and Analysis

Our study focused firstly on the use of existing desktop-based docking tools, as well as the users' impressions on how VEs equipped with multimodal devices might affect their work. Four researchers in bioinformatics accepted to take part in the study. They were aged from 28 to 50 years (M=36, SD=9.95), and all of them routinely used docking programs as part of their everyday work. On the average, researchers had 5 years of experience in the use of docking tools (Min=2, Max=9).

Two techniques were used for data collection: semi-structured interviews and observation-based task analysis.

For the first technique, four interviews were carried out in the workplace and recorded. Confidentiality and anonymity were respected. Interviews were based on a set of questions spanning various topics such as the type of docking problems addressed in day-to-day work, software used, typical difficulties encountered, and how these researchers envisioned working in a VE. Interviews were transcribed *verbatim* and submitted to cognitive discursive analysis using the *Tropes* software program. Analysis was centered on the topic of molecular interaction to identify discourse elements (e.g. data sources, decision criteria) involved in evaluating such interactions.

The second technique consisted in the recording and analysis of three work sessions during which three researchers used 4 standard docking softwares: *ICM-DISCO* [13], *ClusPro* [9], *Hex* [25], and *M-ZDock* [23]. The videotapes served as a basis for our task analysis. Think-aloud protocols [12] were collected throughout and analyzed using explicitly verbalized tasks as a unit of analysis. The coding of actions and verbalizations from observed work sessions provided two complementary supports for the subsequent stage of design : a hierarchical goal decomposition based on the Hierarchical Task Analysis (HTA) methodology [4], and an illustrated summary detailing, by means of a storyboard, the sequences of tasks involved in solving a staple docking problem identified during the preliminary interviews: docking-based modeling of the barnase / barstar complex.

### 3.2 Results

Results allowed us to construct three elements relevant to the design of a multimodal VE for molecular docking: (1) a model of the task carried out, (2) a model of user needs and (3) a model of design principles to be adopted.

#### 3.2.1 Hierarchical Task Analysis

The HTA task tree (see Figure 3) provides a view of how docking activities can be decomposed according to the objectives it underlines. In brief, the first task is to generate large numbers of potential complex models. As input to this task, files detailing the 3D structure of unbound proteins, stored in the Protein Data Bank, are used [6]. The second stage calls on an automatic, manually configured, clustering algorithm to eliminate physically impossible solutions and sort the remaining complexes in clusters based on similarity of properties. In the final stage, remaining complexes are sorted according to a global quality index known as a "score" which can also be configured. Based on this index, the algorithm generates an ordered list of solutions. The user finally selects a small number of results to be tested using experimental methods.

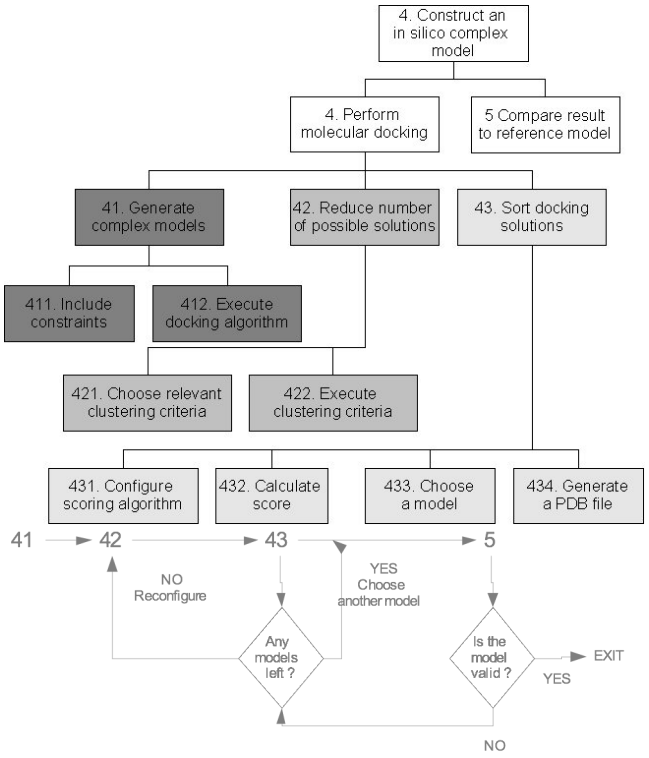



Figure 2. Part of the HTA task tree for molecular docking (top) and task plan (bottom).

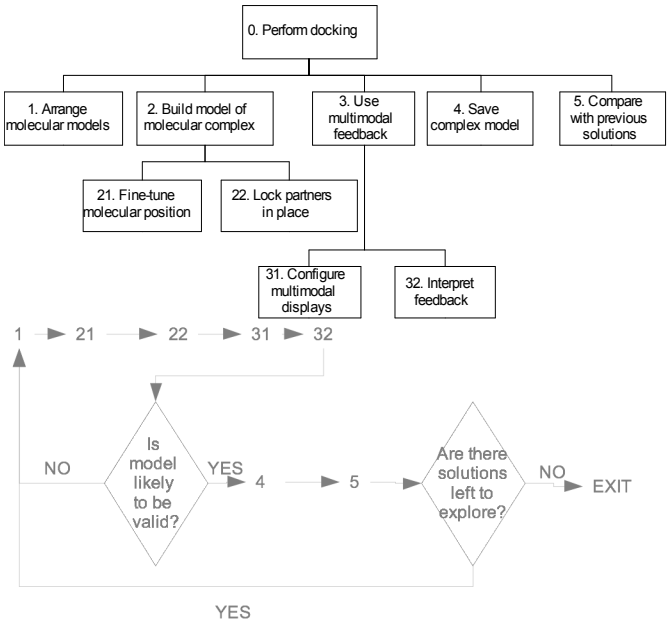



Figure 3. Possible HTA task tree for docking performed under our docking paradigm (top) and task plan (bottom)

Many tasks in Figure 2 consist in a preliminary action on the part of the user, followed by automatic processing carried out by the system, resulting in a situation where application of user knowledge to solve the docking problem is discontinuous and iterative, as evidenced by the task plan.

Exploring the design space resulted in elaborating a possible new way to carry out the docking task in a multimodal VE, depicted in Figure 3 using the same HTA formalism. The first main difference is that possible solutions would be constructed one by one, rather than generated massively and progressively screened. The second one is that the score is not the single, only possible tool to solve a docking problem. This new interaction paradigm implies that the users should be provided with:

- Control over relative position and orientation of both proteins;
- Feedback over the interaction with simultaneous display of all relevant information to the user;
- Support for comparison between several complexes constructed in this fashion.

### 3.2.2 Required displayed information for protein-protein docking task

Four informational needs were initially extracted from the analysis of interview corpora, the first three of which related to molecular properties, and the last to prior knowledge of interface properties.

- Electrical charge can cause attraction or repulsion depending on whether charges bear opposite or same signs, respectively;
- Hydrophobic residues attract each other by repulsing interspersed water molecules;
- Topological factors can cause repulsion in case of collision between parts of the molecules which do not fit together, but cannot cause attraction;
- Prior knowledge of interface properties, specifically which residues ("hot spots") are likely to be involved in the interaction, i.e. to belong to the interface.

Interestingly, references to the score bore no statistical weight in the cognitive-discursive analysis, suggesting that far from being a central notion, the score should rather be considered as a means to an end. Interviews showed that docking involves studying these three molecular properties with respect to prior knowledge of interface properties. Indeed, researchers spent on average 42% of task completion time consulting external sources of data, e.g. the *PubMed* database or personal notes, specifically to search for coordinates of such residues.

### 3.2.3 Inferred allocation of modalities in a VE for protein-protein docking

The term « modal allocation » was introduced by André [3] to describe the use of a specific output modality to display specific information in a multimodal environment. Our goal here is to ensure optimal modal allocation considering both technical (e.g. memory demands) and task-related factors, e.g. depending on the characteristics of the information, on how and when it is used by the researcher.

Despite several decades of work in developing Human Computer Interfaces equipped with multimodal displays, principles for choosing a relevant display modality are rare. Little work focuses on multimodal VE for data exploration [22]. In the case of molecular docking, task analysis determined that at most four types of information need to be displayed at any one time: molecular topology, hydrophobicity, electrical force and residues involved.

Design principles involve giving weight to specific allocations of modalities based on task-related and technical constraints. Task analysis allowed us to identify the following task-related constraints:

*Avoid sensory overload*: Information needs to be balanced across modalities to avoid visual cluttering [20]. Although information sonification and haptization are likely to offload the visual channel and improve performance, arbitrarily assigning information to output modalities might result in decreased performance in comparison to a monomodal setting. Our claim is that taking information semantics into account should allow to avoid this risk. Three aspects, described below, should be taken into account.

*Static vs. Dynamic rendering:* Some information such as molecular structure is necessary at all times. Permanent rendering of this information using auditory or haptic signals might lead to filtering over time or deliberately switching off these functions because they interfere with carrying out the task [20]. The visual modality should therefore be used as a framework in this case.

*Semantics in haptic force feedback*: As we mentioned above, some information must be taken into account in building a protein-protein complex, such as electrostatic forces or hydrophobic interactions for example. Haptic constraining seems a likely choice for electrostatic interaction since it can convey both attraction and repulsion in a way that is congruent with everyday human experience. Visual and auditory modalities should not be discarded immediately, but do require some mapping in which both the semantics and the zero-force point should be immediately apparent (e.g. a dual-tone signal).

| | Visual | Audio | Haptic |
|---|---|---|---|
| **Topology** | | | |
| Protein surface | X | | |
| Collision | X | X | X |
| Surface complementarities | X | X | X |
| **Electrostatic charge** | | | |
| Electrostatic field | X | | X |
| Electrostatic docking score | X | X | |
| **Hydrophobic** | | | |
| Hydrophobic patch | X | | |
| Hydrophobic docking score | X | X | |
| **Prior knowledge (hot spots)** | X | | |

Table 1. Modality allocation space restriction

*Two levels of exploration:* The prospective HTA tree (Figure 3) shows that users expect to carry out molecular docking in two main steps. The first one aims to find a potential interaction zone while examining a protein at the molecular level: it relies on prior knowledge, on topological characteristics, and on the identification of « patches » i.e. large areas of similar electric charge or hydrophobicity, as evidenced by verbal protocols. The second one, at the atomic level (i.e. that of residues) aims to examine compatibility between two potential contact zones at the molecular interface, and relies on a finer examination of the interaction, i.e. electrostatic, hydrophobic and Van der Waals forces. While trying out various solutions, the user needs to access information regarding areas of particular interest (e.g. possible chemical bonds between two molecular areas, hydrophobic patches, electronic field), as well as overall indicators about the complex as a whole (e.g. value of the binding energy).

These constraints allow us to sum up design principles for the four information needs identified in section 3.2.2 (Table 1):

- Display molecular surfaces at least visually to support easy manipulation of molecular models ;
- Display all data involved in the computation of binding energies simultaneously, using a combination of modalities ;
- When possible, stay congruent with familiar sensory experiences and task-related data semantics, e.g. render physical collisions and electrostatic forces with haptic modality;
- Auditory signals can render time-dependent information such as the score;
- Use combinations of modalities to simultaneously render a large quantity of information.

## 4 HOSMoS: An Hybrid Approach for Docking in a Multimodal VE

### 4.1 Combining Immersive and Automatic Approaches for Docking

Our approach, named HOSMoS (Human Oriented Selection of Molecular Specimen), has taken into account constraints and recommendations extracted from the state of the art and from the ergonomic study described above.

By allowing the user to use his knowledge of the complex early on, we hope (1) to allow the user to exploit knowledge and hypotheses about the interaction throughout his work, rather than only while configuring the docking algorithm, and (2) to reduce the risk of "false positives" by allowing the user to "home in" on the best solution rather than choose the "best" alternative in a panel of computer-generated complexes.

Our main goal is to allow the user to act directly on the process. Firstly, the user strongly reduces the search space by selecting a small number of interesting areas on the surface of the proteins whereupon computations are executed. Secondly, the user selects protein complexes that are deemed most probable (see Figure 4). This architecture allows the two levels of examination identified in the ergonomic study: a large scale observation and another level focused on the interface between the two proteins.

For this purpose, our system involves three stages. During the first stage, the multimodal VR framework (Figure 4.I), will allow users to manipulate proteins and place them in interesting configurations. Visual stereoscopy, 3D audio and haptic feedback are designed according to the principles established in section 3.2.3. This first stage will enable a quick reduction of the search space using the abilities and specific knowledge of the user regarding protein-protein docking.

The second stage (Figure 4.II) will be similar to classical automatic processes for protein-protein docking but applied on the restricted space chosen by the user.

Finally, the third stage (Figure 4.III) will allow the user to explore the solution space provided by the previous stage, within the VR framework. During the exploration, the user receives multimodal feedback on the energetic stability of each possible docking configuration. This finally leads to extracting relevant configurations that are then tested in biological experimental conditions.

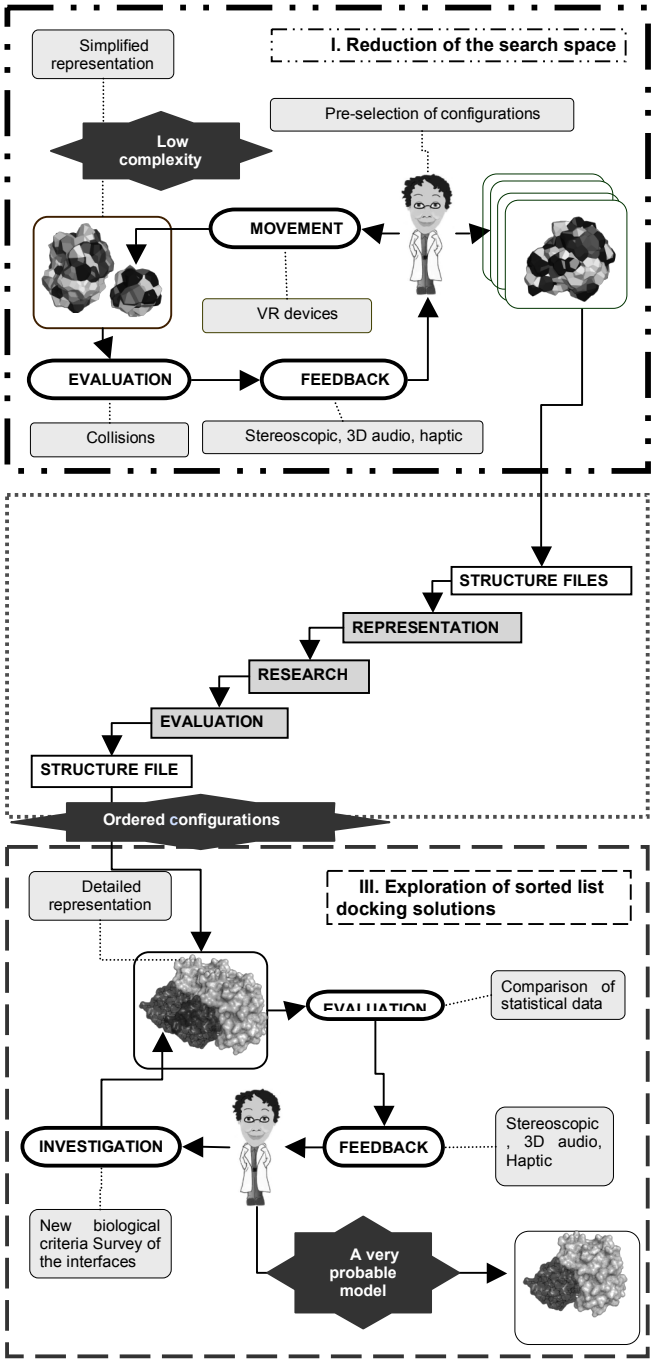



Figure 4. Hybrid HOSMoS approach

## 4.2 HOSMoS Multimodal VE framework

### 4.2.1 Hardware and software architecture

A complete multi-sensory VR platform was available for our study. Active stereoscopy for visual rendering is provided on a two-face CAVE-like [10] system with BARCO® projection technologies and shutter glasses. 3D audio rendering is generated using Max/MSP, spatialised with SPAT, and displayed using 8 loudspeakers or headphones. Moreover, 6 DOF force feedback is provided by a *Virtuose™* device from *Haption®*. Tracking of the user's head for adaptative stereovision and of the hand for 3D Mouse interaction was done by infrared technologies from *ARTrack©*. These devices, as well as gesture and voice recognition systems (to command the application), were managed by a dedicated software platform named *VEserver* [27], which distributes events coming from the devices to the docking application. In our multimodal client/server architecture, each computer deals with a specific modality and stores its own specific 3D representation of the studied proteins. Then they synchronize the position and orientation of protein representations using specific UDP messages (*cf.* Figure 5).

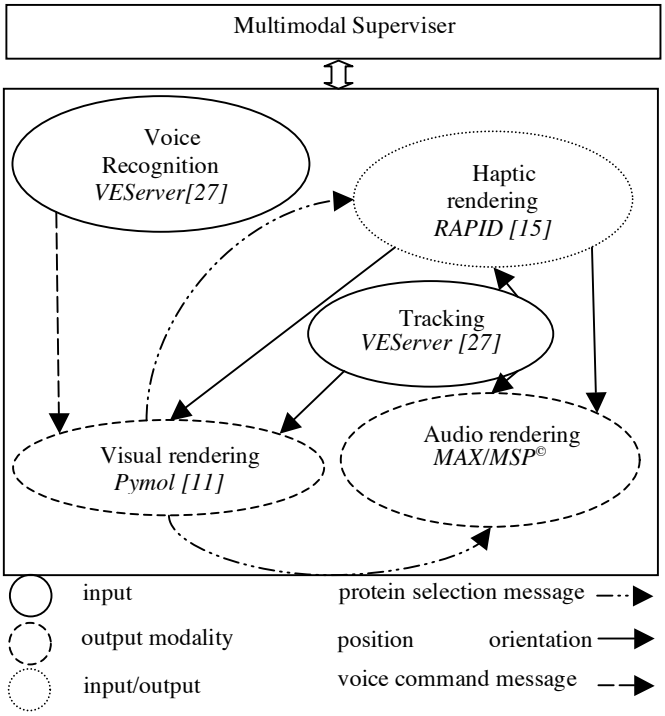



Figure 5. Hardware architecture and synchronization message between computers

### 4.2.2 Multimodal interactions

HOSMoS provides bimanual direct manipulation of two proteins. One is manipulated with a tracked 3D mouse and the other is attached to a 6DoF haptic device as shown in Figure 6.

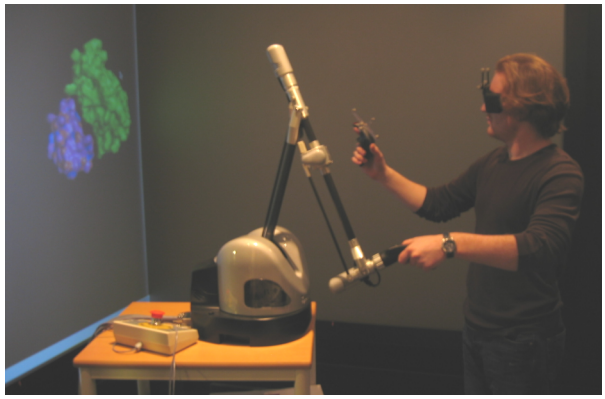


Figure 6. User testing bi-manual haptic docking interaction

#### 4.2.2.1 Visual feedback

Before starting to build a new immersive docking interface, we had to choose a molecular visualization platform that fit with the immersive context and the possible use as expected by biologists. Using an existing platform allowed us to avoid developing existing visual modalities, listed in Figure 4, such as surface rendering or electrostatic patches. The main criteria of our choice were visual rendering efficiency and the ability to easily edit and move molecular object by scripting during visualization. Moreover the visualization platform must be adapted to biologists' standard practices. For these reasons we have chosen the *Pymol* molecular visualization software [11]. This software, often used by biologists, offers *Python scripting API*. This allows the user to use his prior knowledge of the problem at hand (*cf.* Figure 4) within the immersive VR application. This prior knowledge essentially refers to *hotspots*. A *hotspot* is a selection, using the 3D mouse, of atoms on the surface of the protein. This selection is then sent to an automatic algorithm to reduce docking time and the size of the search space. Pymol also allows us to use several types of visual display to render global electrostatic or hydrophobic properties using the visual channel. OpenGL objects can also be drawn for more complex visual feedback, such as electrostatic fields or surface complementarity.

#### 4.2.2.2 Haptic feedback

A major technical constraint of applications using haptic feedback is to maintain a refresh rate between 200 Hz to 1 kHz for the haptic loop. This is essential to guarantee stiff and stable rendering. This entails that computations within the application (collisions, scores, etc.) should be carried out in real time, and time latencies should be reduced as much as possible. For that reason, one of the limits of our study is that only rigid bodies are manipulated, for the time being. With this modality, we chose to render collisions during interactive docking using a "lock-and-key" or "LEGO" paradigm, and to render the global electrostatic field by attractive, repulsive and torque force feedback.

In order to render collisions between proteins, we worked on surface representations of proteins using a triangle mesh. For each protein, the mesh surface is computed using *MSMS* [26] based on files from the Protein Data Bank. With *MSMS* we can apply different settings to increase or decrease surface complexity, according to computing efficiency requirements. We then use the *RAPID* [15] library to detect collisions on the triangle mesh during docking interactions. The result of the collision detection is a list of triangles, needed to compute a repulsion force feedback, which is then sent to the haptic device. The norm of the repulsion force is computed from the number of triangles involved in the collision. The direction of this force comes from the current position of proteins, defined by the center of the proteins involved in the collision, and the last recorded position of the protein without collision.

A global electrostatic feedback is computed with APBS,[5] which allows us to compute a grid for the electrostatic field around the larger protein. Charged amino acids of the second protein are immersed in this electrostatic field. A force is computed for each of them, resulting in a global force and torque for the protein. This method allows us to compute feedback for the resulting global electrostatic force, in linear time, according to the number of charged amino-acids in the smallest protein.

#### 4.2.2.3 Audio feedback

Several global docking evaluation scores cannot be computed with a frame rate between 200Hz and 1 kHz, as required by the haptic loop. Therefore, another modality with a lower frame rate should be chosen to render this score, e.g. audio feedback. For the time being, when the haptic modality is unavailable, we use audio feedback to render collisions, and surface complementarity scores between two proteins. The frequency of beeps increases depending on the surface complementarity score: when the surfaces are complementary, a continuous sound is heard. Several classes of global score such as electrostatic score of the docking configuration are linked to the pitch of this beep.

### 4.2.3 Multimodal supervisor

Rendering should be controlled throughout the docking task. Moreover, mapping rules between modalities and data are not valid in all situations: rendering should depend on the data to be communicated but also on the context of the interaction (defined by a triplet of user, environment and system). That is why our multimodal docking application integrates an intelligent artificial manager to adapt multimodal interaction to various environments, to be flexible to various contexts in which the task might be carried out [7].

This manager, called "Multimodal Supervisor", is based on a representation of human information channels, user objectives, the rendering capabilities of the technical architecture, etc. It is in charge of:

- handling a knowledge base of logical mapping rules between different types of information and available multimodal representations;
- analyzing context thanks to observer modules and balancing mapping rules depending on the situation;
- controlling, replacing, modifying or fine-tuning the rendering if it does not fit to the interaction task and/or the context;
- managing communications with the application and cooperation with the user.

In practice, the role of this supervisor is to enable or disable modalities according to the context of the ongoing VR interaction and user settings, based on messages received and sent from each computer of the VR architecture

## 5 Conclusion

In this paper, we presented a new immersive approach to help biologists in the task of molecular docking. Our immersive application (see section 4) was designed and developed according to in-depth ergonomic study results presented in section 3. The goal of this preliminary study was to determine design principles aiming to integrate VE-based interactions in the everyday practice of researchers in biochemistry, during the docking task. Although efforts have been made to take user needs into account in the design of a novel interface for molecular docking, this approach is in its early stages. We are now entering the stage of evaluating this virtual framework based on user testing. Everyday practice is usually based on an exhaustive automatic approach, and provides a large, ordered set of docking results, which are then manually filtered. Our approach provides an interactive multimodal environment for reducing the search space and consequently decreasing the size of the solutions set. This framework then allows users to filter results according to information rendered by multimodal feedbacks about the various configurations constructed in this fashion. Evaluation must focus on using an iterative process for solving docking problems, in order to show whether using the multimodal VE docking framework is more efficient as regards problem solving time and reliability of results than the existing process.

In addition, biologists use models of rigid proteins such as *pdb* files downloaded from the *Protein Data Bank*. The "lock and key" or "LEGO" metaphor used here is misleading because it suggests all solutions might be reached easily by slotting one protein into the other. However several classes of proteins can adjust their

structure by shifting some of their areas to optimize interaction, a phenomenon known as *induced fit*. In our multimodal VR framework, this would imply that the 3D protein models would be flexible rather than rigid. We are currently studying a new approach for rendering molecular kinematics in real time using the a hardware-based physical engine, SDK, to take into account rigid and flexible areas on the amino-acid chain of the protein.

## REFERENCES


[1] Anastassova, M., Mégard, C. & Burkhardt, J.M.. Prototype evaluation and user-needs analysis in the early design of emerging technologies. In *Proceedings of the 12th International Conference on Human-Computer Interaction*, (Beijing, China: 22-27 July), 2007.

[2] Anderson, A. & Weng, Z. VRDD: applying virtual reality visualization to protein docking and design. *Journal of Molecular Graphics and Modelling,* 17(3), 1999, 180-186.

[3] André, E. The generation of multimedia presentations. in R. Dale, H. Moisl & H. Somers (Eds.), *Handbook of natural language processing : techniques and applications for the processing of language as text*, Marcel Dekker, 2000, 305-327.

[4] Annett, J. Hierarchical Task Analysis, In Hollnagel, E., *Handbook of Cognitive Task Design*, Mawhah NJ: Lawrence Erlbaum, 2003, 17-35.

[5] Baker, NA. Sept, D. Joseph, S. Holst, MJ. McCammon, JA. Electrostatics of nanosystems: application to microtubules and the ribosome. *National. Academy of. Science.*of the USA 98, 10037-10041 2001.

[6] Berman HM, Westbrook J, Feng Z, Gilliland G, Bhat TN, Weissig H, Shindyalov IN, Bourne PE. The Protein Data Bank. *Nucleic Acids Res*. 28(1), 2000, 235-242.

[7] Bouyer, G., Bourdot, P., & Ammi, M. Supervision of Task-Oriented Multimodal Rendering for VR Applications, *Eurographics Symposium on Virtual Environments* (EGVE 2007), (15-18th July 2007, Weimar, Germany), 2007.

[8] Brooks, F. P., Ouh-Young, M., Batter, J. J., & Jerome Kilpatrick, P. Project GROPE: Haptic displays for scientific visualization. *SIGGRAPH Comput. Graph*. 24, (4), 1990, 177-185.

[9] Comeau SR, Gatchell DW, Vajda S, Camacho CJ. ClusPro: an automated docking and discrimination method for the prediction of protein complexes. *Bioinformatics* 20(1), 2004, 45-50.

[10] Cruz-Neira, C. Sandin, D.J. DeFanti, T. A. Kenyon, R. V. Hart, J.C. The CAVE: "Audio Visual Experience Automatic Virtual Environment, Communications of the ACM/SIGGRAPH, 35(6), 1992, 64-72.

[11] DeLano, W.L. The PyMOL *Molecular Graphics System* (2002) on World Wide Web http://www.pymol.org

[12] Ericsson, K. A., & Simon, H. A. *Protocol analysis: Verbal reports as data (revised edition)*. Cambridge, MA: Bradford Books / MIT Press, 1993

[13] Fernandez-Recio J., Totrov M. & Abagyan R. ICM-DISCO docking by global energy optimization with fully flexible Side-Chains. *Proteins*, 52(1), 2003, 113-117.

[14] Garcia-Ruiz M.A. & Guttierez-Pulido J.R. (2006). An overview of auditory display to assist comprehension of molecular information. *Interacting with Computers*, 18(4), 2006, 853-868.

[15] Gottschalk, S. Lin, M. C. and Manocha, D. OBBTree: a hierarchical structure for rapid interference detection. *SIGGRAPH '96: Proceedings of the 23rd annual conference on Computer graphics and interactive techniques (ACM/)*, 1996, 171–180.

[16] Hart, T. N. & Read, R. J., "A multiple-start Monte Carlo docking method", *Proteins: Structure, Function, and Genetics*, 13(3), 2004, 206-222.

[17] Humphrey, W., Dalke, A. and Schulten, K., "VMD - Visual Molecular Dynamics", *Journal of .Molecular. Graphics and Modeling*, vol. 14, 33-38, 1996,

[18] Levine, D. Facello, M. Hallstrom, P. Reeder, G. Walenz, B. & Stevens, F. Stalk: an interactive system for virtual molecular docking. In *Proceedings of IEEE Computational Science and Engineering*, 4(2), 1997, 55-65.

[19] Maciejewski, R., Choi, S., Ebert, D. S. & Tan, H. Z. Multi-Modal Perceptualization of Volumetric Data and Its Application to Molecular Docking. In *Proceedings of the First Joint Eurohaptics Conference and Symposium on Haptic interfaces For Virtual Environment and Teleoperator Systems 0*, 2005.

[20] Mayer, R.E. & Moreno, R. Nine Ways to Reduce Cognitive Load in *Multimedia Learning. Educational Psychologist* 38(1), 2003, 43-52.

[21] Morris, G. M. Goodsell, D. S. Halliday, R. S. Huey, R. Hart, W. E. Belew, R. K. & Olson, A. J. Automated docking using a Lamarckian genetic algorithm and an empirical binding free energy function, *Journal of Computational Chemistry* 19(14), 1999, 1639-1662.

[22] Nesbitt, K.V. (2003). *Designing Multi-sensory Displays of Abstract Data*. Ph.D. thesis, University of Sydney, 2003.

[23] Pierce B, Tong W, Weng Z. M-ZDOCK: a grid-based approach for Cn symmetric multimer docking. *Bioinformatics* 21(8), 2005, 1472-1478.

[24] Ray, N., Cavin, X., Paul, J.C. & Maigret, B. Intersurf: dynamic interface between proteins. *Journal of .Molecular. Graphics and Modeling*. 23(4), 2005, 347-354.

[25] Ritchie, DW. Evaluation of protein docking predictions using Hex 3.1 in CAPRI rounds 1 and 2. *Proteins* 52(1), 2003, 98-106.

[26] Sanner, M. F., Olson A.J. & Spehner, J.-C. Reduced Surface: An Efficient Way to Compute Molecular Surfaces. *Biopolymers* 38, 1996, 305-320.

[27] Touraine, D. Bourdot, P Bellik, Y, Bolot, L. A Framework to Manage Multimodal Fusion of Events for Advanced Interactions within Virtual Environments. In *Proceedings of International EUROGRAPHICS Workshop on Virtual Environment,* (EGVE 2002, Barcelona, Spain), 2002.

[28] Wang, R., Lu, Y. and Wang, S. "Comparative Evaluation of 11 Scoring Functions for Molecular Docking", Journal.of Medicinal. Chemistry, vol. 46, 2003, pp. 287-2303.

[29] Wriggers, W. & Birmanns, S. Interactive fitting augmented by force-feedback and virtual reality. *Journal of. Cellular Biology*, 144, 2003, 123–131.